Mirpur University of Science and Technology
(Mirpur AJK)

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Roll no	**FA23-BSE-054** **FA23-BSE-040** **FA23-BSE-024**
Subject	**Web Engineering**
Submitted to	**Dr. Shamaila Nasreen**
Date	**21/02/2026**

Mirpur University of Science and Technology (Mirpur AJK)

1. Project Title:

Employee Task & Attendance Management System (ETAMS)

2. Problem Understanding & Scenario Fit:

In many organizations, employee attendance and task management is often handled manually using spreadsheets or emails. This creates several issues:

- Inaccurate attendance tracking
- Unclear task assignments
- Missed deadlines
- Difficulty monitoring employee performance

The ETAMS web application solves these problems by providing a centralized platform for:

- Employees to mark attendance and view tasks
- Managers to assign tasks and track progress
- Administrators to manage users and system settings

This leads to enhanced transparency, efficiency, accountability, and productivity in organizations.

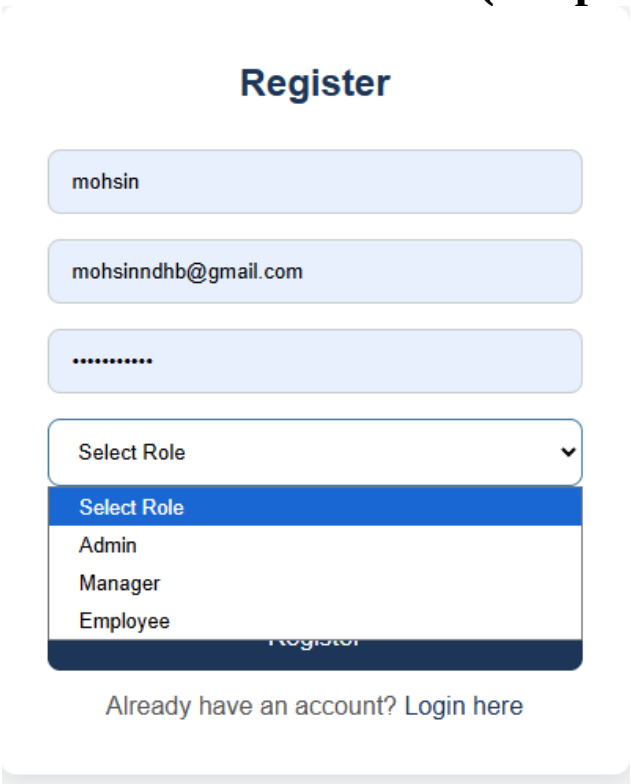
Institution: Mirpur University of Science and Technology (MUST), Mirpur

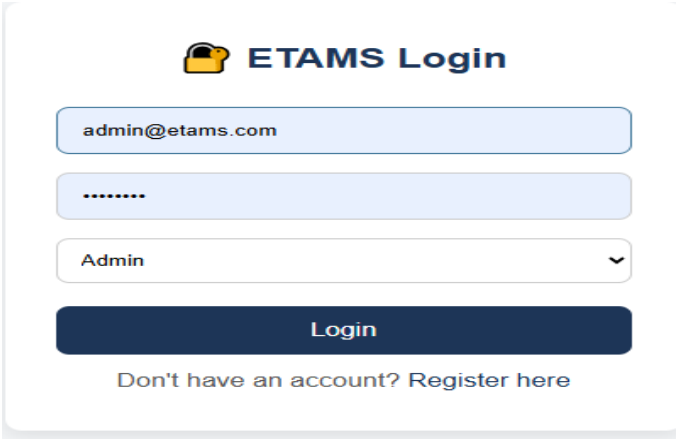
3. Major Features:

User Authentication & Role Management

- Secure login and signup
- Role-based access control (Admin, Manager, Employee)

Output:

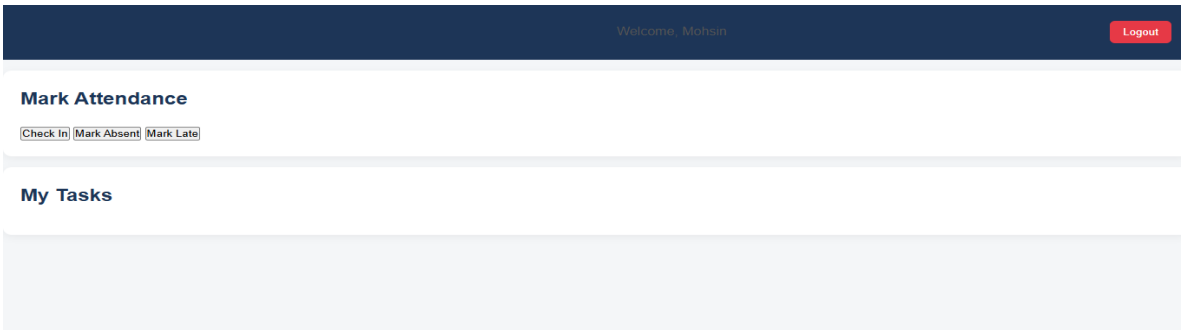

Register
mohsin
mohsinndhb@gmail.com
Select Role
Select Role
Admin
Manager
Employee
Already have an account? Login here


ETAMS Login
admin@etams.com
Admin
Login
Don't have an account? Register here

4. Attendance Management

- Daily check-in and check-out
- Automatic attendance record generation
- Monthly attendance reports

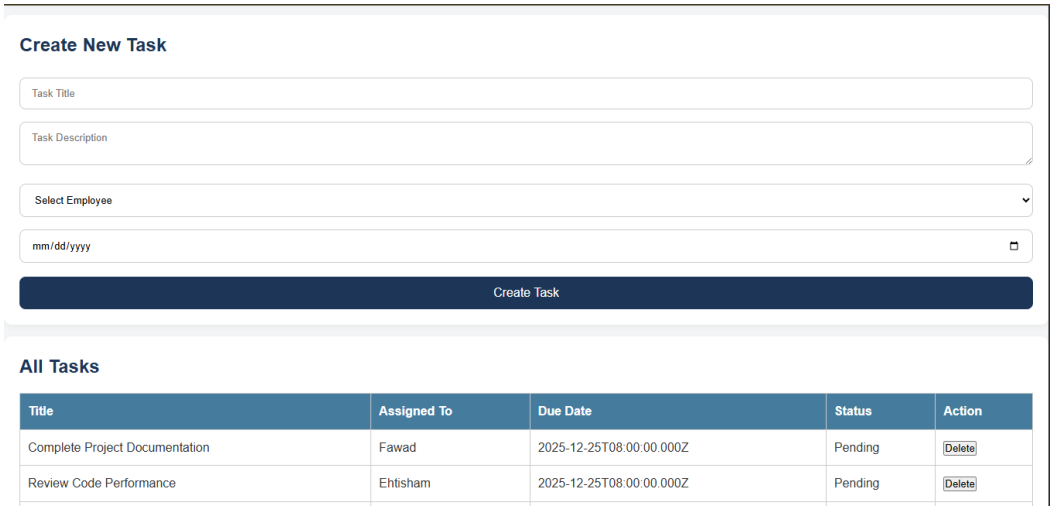
Output:



5. Task Management

- Task assignment by managers
- Task priority and deadline setting
- Task status tracking (Pending, In Progress, Completed)

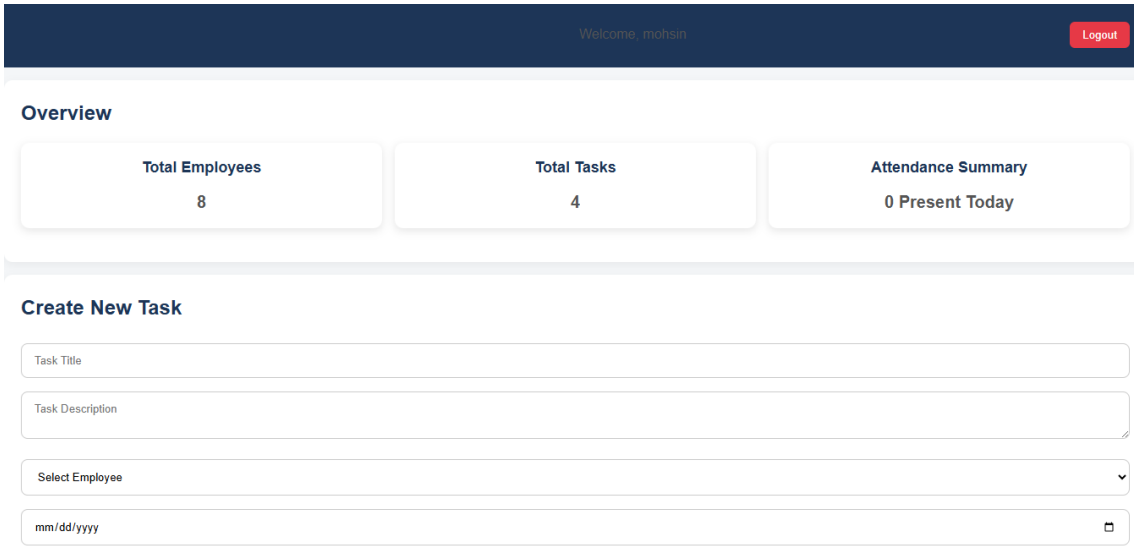
Output:



6. Admin Dashboard

- Total employees
- Attendance overview
- Task statistics
  Purpose: System monitoring

Output:



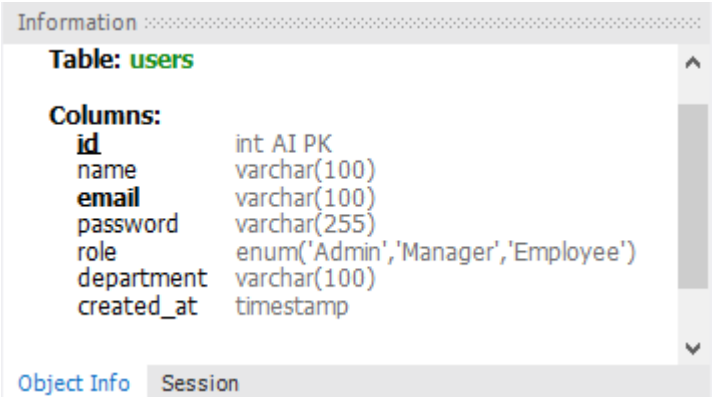
6. Back-End Database Design:

The system uses a relational MySQL database with the following tables:

```
CREATE TABLE users (

   id INT AUTO_INCREMENT PRIMARY KEY,

   name VARCHAR(100) NOT NULL,

   email VARCHAR(100) NOT NULL UNIQUE,

   password VARCHAR(255) NOT NULL,

   role ENUM('Admin','Manager','Employee') NOT NULL,

   department VARCHAR(100),
```

```
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

);
```

Output:



Create attendance table:

```
CREATE TABLE attendance (

    id INT AUTO_INCREMENT PRIMARY KEY,

    user_id INT NOT NULL,

    date DATE NOT NULL,

    status ENUM('Present','Absent','Late') DEFAULT 'Present',

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE KEY unique_attendance (user_id, date),

    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE

);
```

Output:

Information

Table: attendance

Columns:

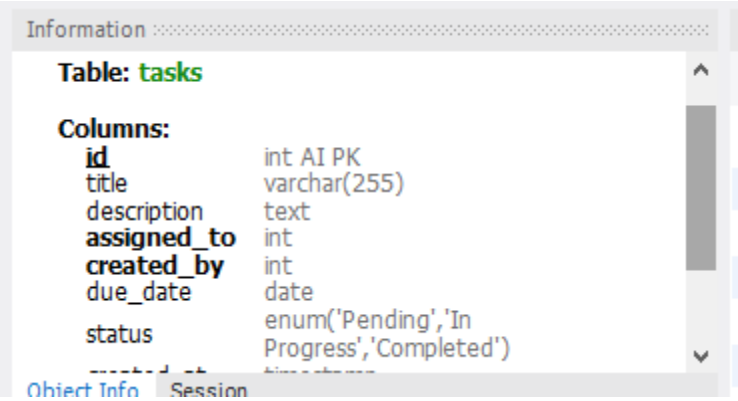
id	int AI PK
user_id	int
date	date
status	enum('Present','Absent','Late')
created_at	timestamp

Object Info Session

Create Task table:

```
CREATE TABLE tasks (
   id INT AUTO_INCREMENT PRIMARY KEY,
   title VARCHAR(255) NOT NULL,
   description TEXT,
   assigned_to INT NOT NULL,
   created_by INT NOT NULL,
   due_date DATE,
   status ENUM('Pending','In Progress','Completed') DEFAULT 'Pending',
   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
   FOREIGN KEY (assigned_to) REFERENCES users(id) ON DELETE CASCADE,
   FOREIGN KEY (created_by) REFERENCES users(id) ON DELETE CASCADE
);
```

Output:



7. Technology Stack:

Front-End: HTML, CSS, JavaScript
Back-End: Node.js, Express.js
Database: MySQL
Authentication: JWT & bcrypt
Middleware: Role-based access control

8. System Workflow:

1. **Login / Signup**: User registers or logs in. JWT token is generated.
2. **Admin**: Can view all employees, attendance, tasks, assign tasks, delete tasks.
3. **Manager**: Can assign tasks, view their created tasks, monitor employee progress.
4. **Employee**: Can mark attendance, view assigned tasks, update task status.
5. **Reports**: Admin and Managers can view task and attendance reports.

9. Key Features Highlight

- **JWT Authentication & bcrypt** ensures secure login
- **Role-based access** prevents unauthorized actions
- **CRUD operations** for tasks and attendance
- **Real-time dashboards** for Admin, Manager, and Employee

10. Conclusion

The Employee Task & Attendance Management System **is a** practical, real-world web application that demonstrates:

- Clear problem understanding
- Structured feature breakdown
- Logical front-end workflows
- Proper database design and backend implementation
- Security using JWT and bcrypt

This system improves efficiency, accountability, and performance tracking for organizations

Github Link:

https://github.com/rizwansemust-maker/ETAMS